

Lab 12 - Fundamental Network Topologies



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Lattes

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Scopus

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substationworm

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Problem Overview

Containerized Hosts

- single-pc1: 172.30.20.10, 02:12:20:00:00:01.
- single-pc2: 172.30.20.11, 02:12:20:00:00:02.
- single-pc3: 172.30.20.12, 02:12:20:00:00:03.
- ring-node1: 172.30.40.2, 172.30.40.34, 02:14:40:00:00:01.
- ring-node2: 172.30.40.3, 172.30.40.10, 02:14:40:00:00:02.
- ring-node3: 172.30.40.11, 172.30.40.18, 02:14:40:00:00:03.
- ring-node4: 172.30.40.19, 172.30.40.26, 02:14:40:00:00:04.
- ring-node5: 172.30.40.27, 172.30.40.35, 02:14:40:00:00:05.
- star-core: 172.30.30.2, 172.30.30.10, 172.30.30.18, 02:13:30:00:00:00.
- star-leaf1: 172.30.30.3, 02:13:30:00:00:01.
- star-leaf2: 172.30.30.11, 02:13:30:00:00:02.
- star-leaf3: 172.30.30.19, 02:13:30:00:00:03.

Networks

- single-net: 172.30.20.0/24.
- ring-ab: 172.30.40.0/29.
- ring-bc: 172.30.40.8/29.
- ring-cd: 172.30.40.16/29.
- ring-de: 172.30.40.24/29.
- ring-ea: 172.30.40.32/29.
- star-net-a: 172.30.30.0/29.
- star-net-b: 172.30.30.8/29.
- star-net-c: 172.30.30.16/29.

Note: The single-net is defined as a simple, flat Layer 2 network composed of a single Ethernet segment. The ring topology, consisting of the segments ring-ab, ring-bc, ring-cd, ring-de, and ring-ea, is conceptual and implemented using static routes only; therefore, no reconvergence occurs in the event of a link failure. The star topology comprises a central node (star-core) operating as a router and three connected subnets (star-net-a, star-net-b, and star-net-c).

Tasks

- **1** Access single-pc1 using the command `./OOTab12 -run single1` and list the ARP cache by executing `arp -a`. Then, access single-pc2 and single-pc3 in two additional terminal sessions. On both single-pc2 and single-pc3 hosts, initiate ICMP traffic monitoring using `tcpdump -i any icmp`. From single-pc1, perform a ping sweep across the single-net network: `fping -a -g`. After completing the sweep, run again `arp -a`. Analyze and discuss the changes observed in the ARP cache.
- **2** On single-pc1, execute a ping scan using `nmap -sn`. During this scan, capture ARP traffic on single-pc2 and single-pc3 using: `tcpdump -i any arp`. Identify and report which hosts are detected as up.
- **3** Access ring-node1 using: `./OOTab12 -run ring1`. Evaluate the logical paths to ring-node2, ring-node3, ring-node4, and ring-node5 using: `traceroute`. Map the routes, evaluate the hop sequences, and draw a network topology diagram.
- **4** Disable one logical interface on ring-node1: `ip link set eth1 down`. Repeat the traceroute procedure to ring-node2, ring-node3, ring-node4, and ring-node5. Analyze the impact of the interface disruption on path continuity. Note that the OTLab12 uses static routing and no automatic failover is implemented. In contrast, real-world ring topologies typically rely on dynamic routing protocols or redundancy mechanisms.
- **5** Access star-leaf1: `./OOTab12 -run star1`. Test connectivity by sending ICMP echo requests to star-leaf2 and star-leaf3. Map the routes, evaluate the hop sequences, and draw a network topology diagram.
- **6** Restart OTLab12 to restore all hosts to their default configuration. In separate terminals for single-pc2 and single-pc3, monitor ARP traffic: `tcpdump -i any arp`. On single-pc1, generate ARP broadcasts: `arping -c 3 172.30.20.11`. Document the results and identify the broadcast domains.
- **7** In separate terminals for ring-node2 and ring-node3, monitor ARP traffic: `tcpdump -i any arp`. From ring-node1, execute: `arping -c 3 172.30.40.3`. Document the results and identify the broadcast domains.
- **8** In separate terminals for star-leaf2 and star-leaf3, monitor ARP traffic: `tcpdump -i any arp`. From star-leaf1, execute: `arping -c 3 172.30.30.2`. Document the results and identify the broadcast domains.
- **9** Use the arp-scan utility to validate and confirm broadcast domains in each of the three network scenarios implemented in OTLab12.
- **10** For each of the three OTLab12 network scenarios, map the Layer 3 topology using: `nmap -sn --traceroute <sequential_list_of_all_IPAddresses>`. Analyze the resulting routing paths and compare them with the expected logical topology. Depending on the responses observed during host discovery and topology reconstruction, the following alternative command may be helpful for your analysis: `nmap -sn -PE -vv --packet-trace <sequential_list_of_all_IPAddresses>`.

Tools

- The following tools are available for completing OTLab 12: arp, arp-scan, arping, fping, nmap, ping, tcpdump, and traceroute.

Nomenclature

- ARP: Address resolution protocol.
- ICMP: Internet control message protocol.
- IP: Internet protocol.